



What happened to Cap?

The directors and writers do not seem to agree on what happened to Cap in Endgame. <https://dgit.in/june19-3>



The Möbius Time Loop

Check out this Möbius strip template for the characters of Endgame. <https://dgit.in/june19-4>

Van Winkle jumps through this wormhole, he would have travelled 20 years into the past. This is how you can build a time machine, without violating the laws of physics. A solution to Einstein's field equations called the Godel Metric, allows time (or what physicists call world lines) to bend in on itself, providing a theoretical basis for time travel. The hypothetical Alcubierre drive does not move faster than light, but contracts space in front of it and expands space behind it, allowing for faster than light travel. A spaceship using such a drive could as easily be a timeship. Another way to do it is to have an infinitely long cylinder that spins really fast, dragging spacetime around it till it warps into itself. Such a cylinder is called the Tippler Cylinder. The routes of these hypothetical time travels are called closed timelike curves or CTCs. Stephen Hawking proposed that the laws of physics do not allow such CTCs to exist, except on submicroscopic scales, in his Chronology protection conjecture. Essentially, this prevents you from going back and killing your own grandfather, causing the famous paradox that prevents you from existing in the first place. In Endgame however, Thanos dies in 2004 before he can cause the snap. So, what gives? As The Hulk puts it, "time does not work this way".

THE INVERTED MOBIUS STRIP

This is where the inverted Möbius strip comes into play. Stark checks with Jarvis if time travel is possible, and finds a way that it is. A Russian physicist by the name of Igor Novikov has suggested that time is consistent in

itself, as if someone went back in time and took an action, the consequences have already played out without any paradoxes. This is not how time travel works in Endgame though. While War Machine, Ronin and Ant Man do not go back in time to kill Baby Thanos, Nebula does kill the 2014 version of herself, which should not be possible. The Ancient One visually shows the "many worlds" interpretation to the Hulk, showing the universe splitting into an alternative timeline. David Deutsch, an English physicist came up with a solution to the grandfather paradox, which explains all eventualities as probabilities. In a double slit experiment, the universe itself branches into two, and the photon goes through one slit in each universe. When you open up a box with Schrodinger's cat within it, in one universe the cat is dead, while in another universe, the cat is alive. In the context of Endgame, there is $\frac{1}{2}$ a chance of the time heist taking place, $\frac{1}{2}$ chance of Thanos snapping the universe out of existence, $\frac{1}{2}$ a chance of Thanos dying in 2014, $\frac{1}{2}$ a chance of Thor becoming Lebowski Thor and so on. It all fits beautifully, and the scientific paper is called "Quantum Mechanics Near Closed Timelike Lines". Both these sets of eventualities take place, in a CTC that bends in on itself. The good news is that there is a version of the timeline where Stark and Nat do not die, Ant Man is never stuck in the quantum realm, and Thanos never gets to kill half the universe in the first place. While the explanation works, even if you plot out the storylines and build your own Möbius strip, like the way Stark did with

his volumetric display, the technical term used for it is made up.

THE BALDERDASH

Being the big budget Hollywood movie for the masses that it is, there is a bunch of scientific gibberish thrown in, mostly spouted by Stark. When describing nature with waves instead of particles, you need a different vocabulary to measure parameters such as energy. These measurements are called eigenvalues, and are associated with eigenfunction. Each parameter has an associated operator, and for energy, it is called a Hamiltonian. The energy of an electron is its eigenvalue. Finding out all the possible eigenvalues of a particle, is called spectral decomposition, or eigendecomposition. The Planck scale just deals with the smallest possible spaces known to science, beyond which known physics fails to function as expected. The units include Planck length and Planck time. Just after the birth of the universe at the big bang, everything was contained within the Planck scale. Quantum fluctuations are just temporary changes in the energy levels of space at a given point, and take place constantly. They are not as destabilising or dangerous to reality as Stark makes it out to be. Armed with this information, it is easy to understand sentences such as "Quantum fluctuation messes with the Planck's scale, which then triggers the Deutsch Proposition". There is no such thing as the "Deutsch Proposition" that Stark mentions. However, Deutsch Prepositions are a thing, which might be an obscure linguistics/grammar joke. If you get it, let us know. [1](#)



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